01

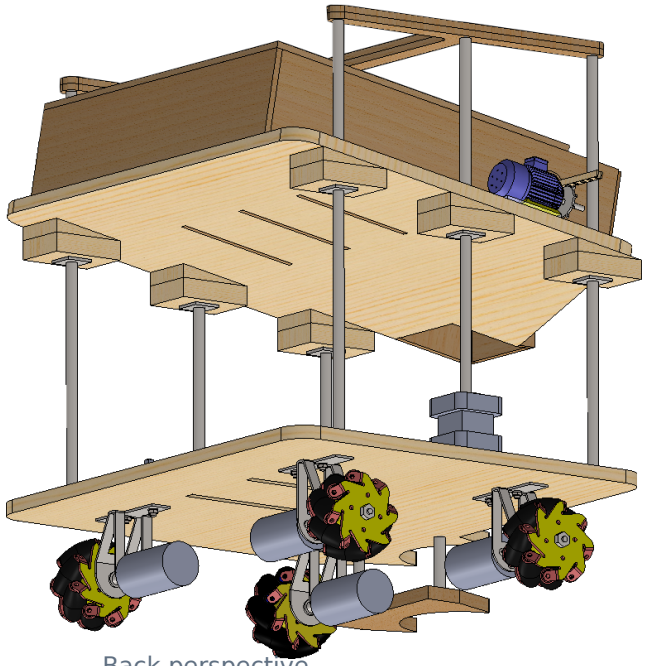
WARMAN 2020 | PROJECT FLAMEOUT

Engineering design challenge - collaborative four-person coursework project

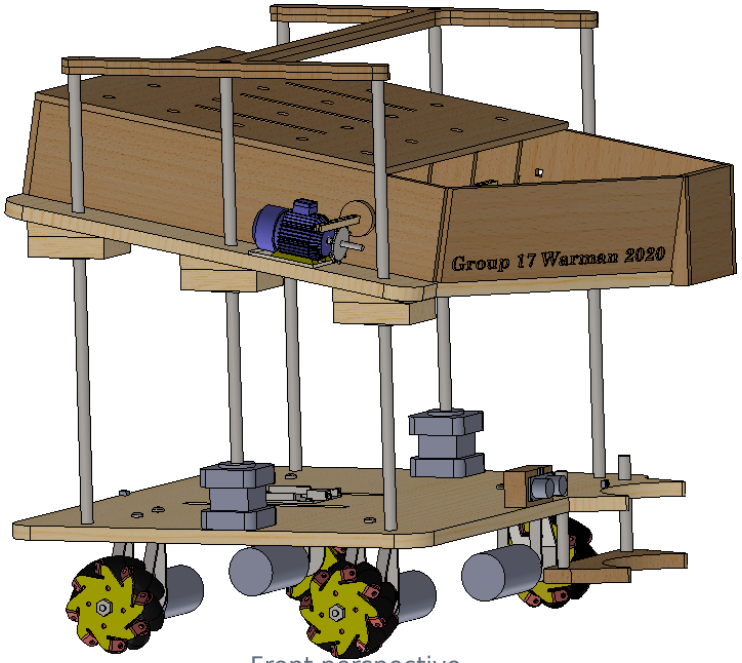
THE BRIEF

Design a conceptual mobile device to transport and deposit 10 tennis-ball payloads into target well-head tubes on a prescribed track, within competition constraints. The 2020 project was evaluated through design and simulation, not physical trials.

FINAL DESIGN / CAD RENDERS



Back perspective



Front perspective

02

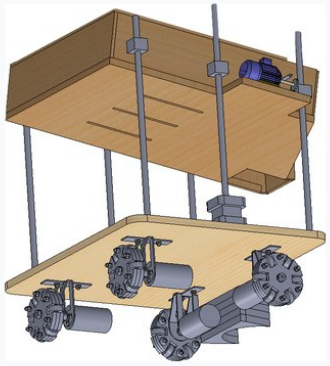
DESIGN EVOLUTION

Three CAD iterations from the team report - back and front views

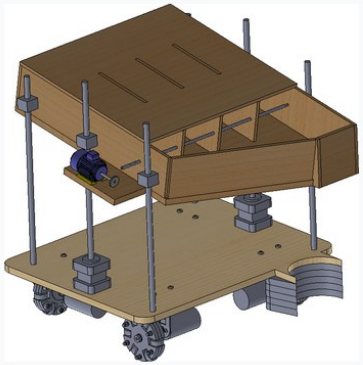
DESIGN ITERATIONS

Warman 2020 / Group 17 / SolidWorks CAD concepts

Iteration 1

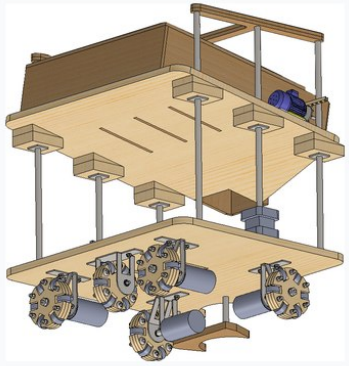


Back view

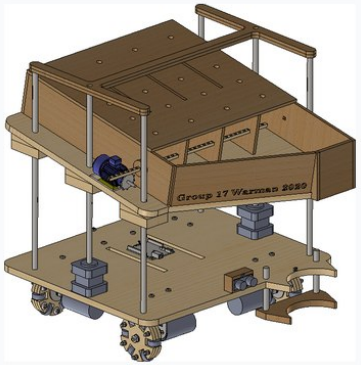


Front view

Iteration 2

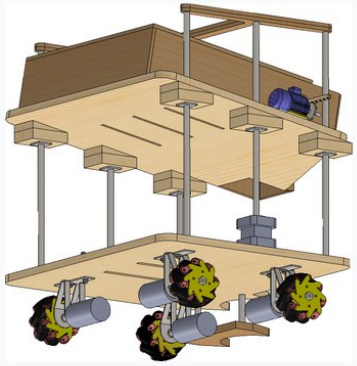


Back view

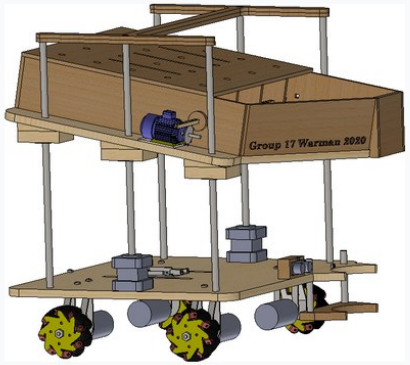


Front view

Iteration 3



Back view



Front view

Conceptual engineering designs; no physical prototype was built in 2020.

03

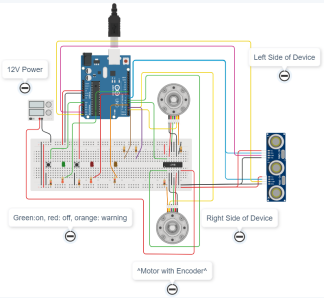
ENGINEERING WORKFLOW & OUTCOME

Selected technical evidence from the original team coursework

DESIGN PROCESS

1. Generated and compared multiple embodiment concepts.
2. Used morphological analysis and weighted decision tables.
3. Selected a gravity-assisted roll-and-drop approach.
4. Developed CAD assemblies and revised the wheel/chassis.
5. Produced a kinematic simulation and controller concept.

CONTROL CONCEPT



Arduino, motors, encoders and sensing concept

SCOPE & LIMITATIONS

The team report estimated a 70-second run and a score of 145/180 under assumed perfect ball delivery.

These are projected simulation-based figures, not an experimentally measured competition result.

The report explicitly states the device could not be physically constructed or tested in 2020.

CREDIT & SOURCE

Selected images and descriptions adapted from the collaborative Warman 2020 Group 17 coursework report.

Tianyuan (Chris) Liu was a member of the four-person team; the documented design is a team outcome.

The original report references externally sourced wheel geometry; images here retain that design context.

This is a portfolio excerpt, not the complete academic submission or an official competition certificate.